

DATA SCIENCE INTERNSHIP PROGRAMME

Week 2: Data Preprocessing Report

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Loan Prediction Dataset

Feature Engineering & Data Preprocessing for Machine Learning

Item	Details
Programme	Data Science Internship Programme
Week	Week 2
Dataset	Loan Prediction Dataset
Target	Loan_Status
Final size	613 rows × 13 columns
Final status	Machine-learning ready

1. Introduction

This report documents the data preprocessing and feature engineering workflow applied to the Loan Prediction dataset for Week 2. The objective was to transform the raw business dataset into a clean, consistent numerical dataset suitable for subsequent machine-learning modelling.

The workflow covered dataset inspection, data cleaning, feature engineering, categorical encoding, outlier detection and treatment, feature selection, numerical scaling, final validation, and export.

2. Dataset Inspection

The dataset was loaded with Pandas and inspected before preprocessing. The structure, data types, missing values, duplicate records, and summary statistics were examined. The original dataset contained 614 application records; after the final duplicate removal, the processed dataset contained 613 unique observations.

- Displayed the first rows of the dataset.
- Examined rows and columns and reviewed data types.
- Assessed missing values.
- Checked duplicate records.
- Reviewed summary statistics for numerical variables.

3. Data Cleaning

3.1 Missing Values

Missing values were assessed and handled during preprocessing. The final validation confirmed 0 missing values across all 13 final columns.

3.2 Duplicate Records

The final readiness check initially identified one duplicate record. It was removed before final validation. The final dataset therefore contains 613 rows and 0 duplicate records.

3.3 Data Types

Categorical information was converted into numerical representations. Final validation confirmed that no non-numerical/object columns remained.

4. Feature Engineering

TotalIncome was created by combining ApplicantIncome and CoapplicantIncome. This provides a combined income measure for each loan application.

Correlation analysis later showed a strong relationship between ApplicantIncome and TotalIncome (approximately 0.80). ApplicantIncome was therefore removed during feature selection while TotalIncome was retained.

5. Categorical Feature Encoding

Categorical variables were converted into numerical representations. Binary variables were represented numerically, while Property_Area was one-hot encoded into indicator features.

- Gender
- Married
- Dependents
- Education
- Self_Employed
- Property_Area_Semiurban
- Property_Area_Urban
- Loan_Status (target)

One-hot encoding was appropriate for Property_Area because its categories do not have a natural numerical ranking.

6. Outlier Detection and Treatment

Potential outliers were identified using the Interquartile Range (IQR) method. For each selected numerical variable, Q1, Q3 and IQR were calculated, with the conventional bounds $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.

Feature	Potential Outliers	Percentage
ApplicantIncome	50	8.14%
CoapplicantIncome	18	2.93%
LoanAmount	41	6.68%
Loan_Amount_Term	88	14.33%
TotalIncome	50	8.14%

Loan_Amount_Term was excluded from IQR capping because it is a discrete loan-term feature with concentrated legitimate categories. Treating it as a continuous outlier variable could incorrectly alter valid loan terms.

IQR capping was applied only to the continuous financial variables ApplicantIncome, CoapplicantIncome, LoanAmount, and TotalIncome. Capping reduced the influence of extreme financial values while preserving the application records.

A post-treatment boxplot was used to review the distributions after capping.

7. Correlation Analysis and Feature Selection

7.1 Relationship with Loan_Status

Credit_History had the strongest individual relationship with Loan_Status, with a correlation of approximately 0.5406. Property_Area_Semiurban followed at approximately 0.1365, while most other individual correlations were relatively weak.

Feature	Correlation
Credit_History	0.540556
Property_Area_Semiurban	0.136540
Married	0.091478
Education	0.085884
Gender	0.017987
CoapplicantIncome	0.011983
Dependents	0.010118
TotalIncome	0.002220
ApplicantIncome	-0.000442
Self_Employed	-0.003700
Loan_Amount_Term	-0.022549
Property_Area_Urban	-0.043621
LoanAmount	-0.047262

Weak Pearson correlation does not necessarily mean a feature is useless because machine-learning models can capture nonlinear relationships and interactions. Feature selection therefore considers both statistical evidence and business relevance.

7.2 Highly Correlated Independent Features

Using an absolute correlation threshold of 0.70, ApplicantIncome and TotalIncome were identified as strongly correlated, with a correlation of approximately 0.7956.

Because TotalIncome was engineered from ApplicantIncome and CoapplicantIncome, retaining both would introduce redundant information. ApplicantIncome was removed while TotalIncome was retained. The one-hot encoded Property_Area variables were retained because they represent distinct categories.

8. Feature Scaling

StandardScaler was applied to CoapplicantIncome, LoanAmount, and TotalIncome. These continuous financial variables have different numerical ranges, so standardization places them on a comparable scale and is useful for algorithms sensitive to feature magnitude.

Scaled feature	Mean after scaling	Std. deviation
CoapplicantIncome	≈ 0	≈ 1
LoanAmount	≈ 0	≈ 1
TotalIncome	≈ 0	≈ 1

The notebook validation produced means extremely close to zero and standard deviations of approximately 1.0008 for all three scaled variables, confirming successful standardization.

9. Final Feature Set

After cleaning, engineering, encoding, outlier treatment, correlation analysis, and feature selection, the final dataset contains 13 numerical columns:

- Gender
- Married
- Dependents
- Education
- Self_Employed
- CoapplicantIncome
- LoanAmount
- Loan_Amount_Term
- Credit_History
- Loan_Status
- TotalIncome
- Property_Area_Semiurban
- Property_Area_Urban

ApplicantIncome was removed because of its strong redundancy with TotalIncome. Loan_Status was retained as the target variable.

10. Final Machine-Learning Readiness Validation

An automated validation check was performed before export. The final results were:

Validation check	Result
Rows	613
Columns	13
Missing values	0
Duplicate records	0
Non-numerical columns	0
Loan_Status target present	Yes
Machine-learning ready	Yes

The readiness condition was satisfied because there were no missing values, no duplicate records, no non-numerical columns, and the Loan_Status target variable was present.

11. Export of Machine-Learning-Ready Dataset

The validated dataset was exported as a CSV file for subsequent predictive modelling. The notebook confirmed successful saving and a final shape of (613, 13).

12. Overall Preprocessing Summary

Step	Action / Result
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Data inspection	Reviewed structure, data types, missing values, duplicates, and summary statistics.
Data cleaning	Handled missing values and removed duplicate records.
Feature engineering	Created TotalIncome from applicant and coapplicant income.
Encoding	Converted categorical variables to numerical representations and one-hot encoded Property_Area.
Outlier detection	Used the IQR method.
Outlier treatment	Applied IQR capping to continuous financial variables only.
Feature selection	Removed ApplicantIncome because of strong correlation with TotalIncome.
Scaling	Standardized CoapplicantIncome, LoanAmount, and TotalIncome using StandardScaler.
Validation	Confirmed 613 rows, 13 columns, zero missing values, zero duplicates, no object columns, and target presence.
Export	Saved the final machine-learning-ready dataset as CSV.

13. Conclusion

The Week 2 preprocessing workflow successfully transformed the Loan Prediction dataset into a machine-learning-ready format. The process addressed missing values, duplicate records, categorical variables, extreme financial values, redundant features, and differences in numerical scale.

The final dataset contains 613 rows and 13 numerical columns, including Loan_Status as the target variable. Final validation confirmed zero missing values, zero duplicate records, and no non-numerical columns. The dataset is therefore ready for the next stage: predictive machine-learning model development and evaluation.